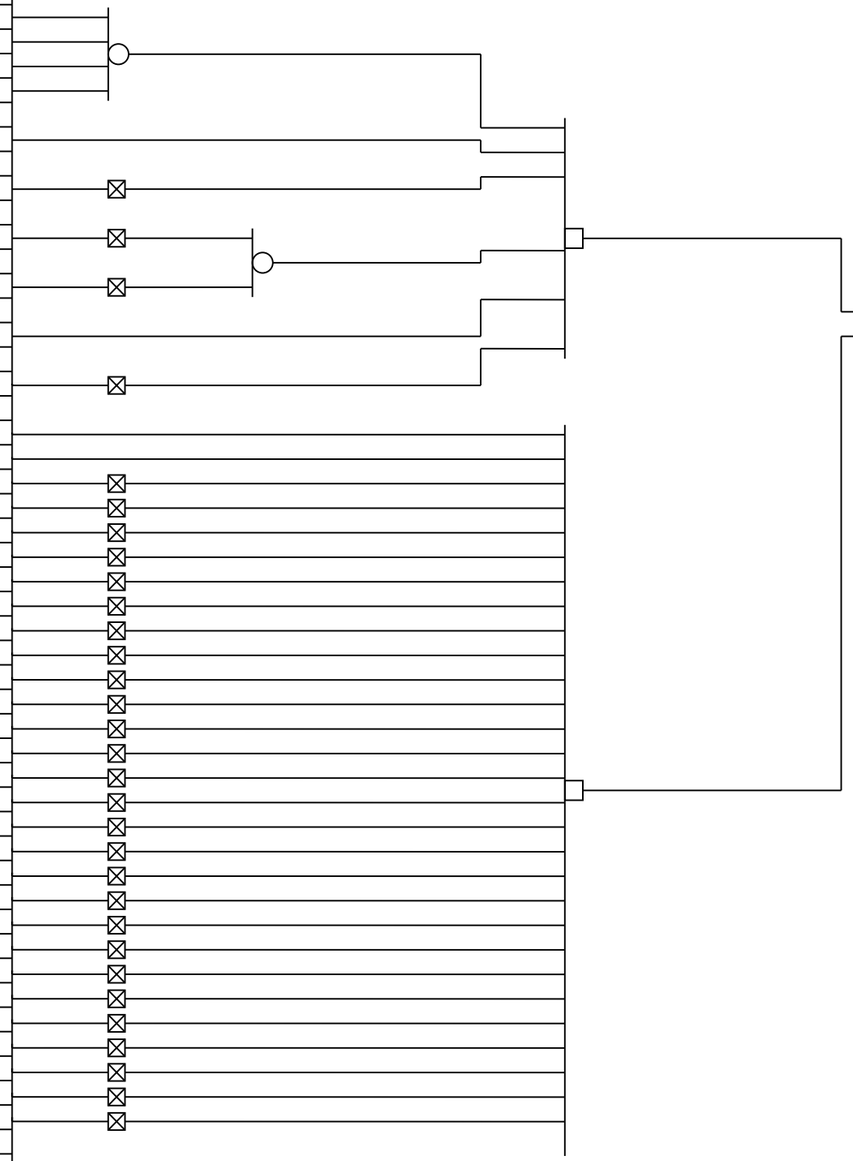
FROM	NO.	TAG NO.	SERVICE
	1		
	2		
	3		
	4		
	5		
	6		
	7		
FG_G-501-89	8		FG WBH INL B/P MOV OPND
FG_G-541-92	9		FG WBH A INL MOV OPND
FG_G-561-92	10		FG WBH B INL MOV OPND
FG_G-581-92	11		FG WBH C INL MOV OPND
	12		
FG_G-201-61	13		FG COMPR SYS REM & RDY TO STR
	14		
FG_G-301-64	15		FG MET CAB FLT
	16		
FG_G-301-69	17		FG MET CAB FC#1 FLT
	18		
FG_G-301-74	19		FG MET CAB FC#2 FLT
	20		
FG_G-401-64	21		FGS ESD V/V POSN OPN
	22		
FG_G-301-84	23		FG MET CAB FAN FLT
	24		
HEATER LCP	25		WATER BATH HEATER READY TO START
FG_P-001-82	26		FGS ESD V/V INL PRESS > 25 barg
FG_P-005-67	27		FG COM COND DRN TNK LVL > H
FG_P-002-86	28		FG FLTR SEP A UPPER LVL > H
FG_P-002-108	29		FG FLTR SEP A DOWN LVL > H
FG_P-003-86	30		FG FLTR SEP B UPPER LVL > H
FG_P-003-108	31		FG FLTR SEP B DOWN LVL > H
FG_P-004-86	32		FG FLTR SEP C UPPER LVL > H
FG_P-004-108	33		FG FLTR SEP C DOWN LVL > H
FG_P-008-89	34		GTG #11 FG FNL FLTR A UPPER LVL > H
FG_P-008-109	35		GTG #11 FG FNL FLTR A DOWN LVL > H
FG_P-009-89	36		GTG #11 FG FNL FLTR B UPPER LVL > H
FG_P-009-109	37		GTG #11 FG FNL FLTR B DOWN LVL > H
FG_P-011-89	38		GTG #12 FG FNL FLTR A UPPER LVL > H
FG_P-011-109	39		GTG #12 FG FNL FLTR A DOWN LVL > H
FG_P-012-89	40		GTG #12 FG FNL FLTR B UPPER LVL > H
FG_P-012-109	41		GTG #12 FG FNL FLTR B DOWN LVL > H
FG_P-014-89	42		GTG #13 FG FNL FLTR A UPPER LVL > H
FG_P-014-109	43		GTG #13 FG FNL FLTR A DOWN LVL > H
FG_P-015-89	44		GTG #13 FG FNL FLTR B UPPER LVL > H
FG_P-015-109	45		GTG #13 FG FNL FLTR B DOWN LVL > H
FG_P-017-89	46		GTG #14 FG FNL FLTR A UPPER LVL > H
FG_P-017-109	47		GTG #14 FG FNL FLTR A DOWN LVL > H
FG_P-018-89	48		GTG #14 FG FNL FLTR B UPPER LVL > H
FG_P-018-109	49		GTG #14 FG FNL FLTR B DOWN LVL > H
FG_P-010-68	50		GTG #11 FG COND DRN TNK LVL > H
FG_P-013-68	51		GTG #12 FG COND DRN TNK LVL > H
FG_P-016-68	52		GTG #13 FG COND DRN TNK LVL > H
FG_P-019-68	53		GTG #14 FG COND DRN TNK LVL > H
	54		
	55		
	56		
	57		
	58		
	59		

LOGIC



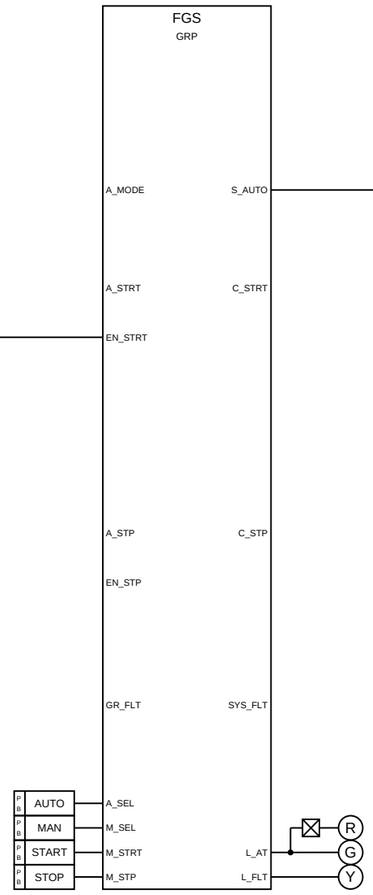
SERVICE	TAG NO.	NO.	TO
		60	
		61	
		62	
		63	
		64	
		65	
		66	
		67	
		68	
		69	
		70	
		71	
		72	
		73	
		74	
		75	
		76	
		77	
		78	
FUEL GAS SYSTEM RDY		79	FG_G-102-17
		80	
		81	
		82	
		83	
		84	
		85	
		86	
		87	
		88	
		89	
		90	
		91	
		92	
		93	
		94	
		95	
		96	
		97	
		98	
		99	
		100	
		101	
		102	
		103	
		104	
		105	
		106	
		107	
		108	
		109	
		110	
		111	
		112	
		113	
		114	
		115	
		116	
		117	
		118	

NOTE :

PROJECT NAME	AL DHAFRA POWER PLANT		
DRAWING TITLE	CONTROL LOGIC & LOOP DIAGRAM FUEL GAS SYSTEM READY		
PROJECT DWG NO. 1A5J-00GEN00-J09-0002		SHEET FG_G-101	REV. 2
PROJECT NO. 1A5J		SCALE NONE	REF. CODE -

FROM	NO.	TAG NO.	SERVICE
	1		
	2		
	3		
	4		
	5		
	6		
	7		
	8		
	9		
	10		
	11		
	12		
	13		
	14		
	15		
	16		
FG_G-101-79	17		FUEL GAS SYSTEM RDY
	18		
	19		
	20		
	21		
	22		
	23		
	24		
	25		
	26		
	27		
	28		
	29		
	30		
	31		
	32		
	33		
	34		
	35		
	36		
	37		
	38		
	39		
	40		
	41		
	42		
	43		
	44		
	45		
	46		
	47		
	48		
	49		
	50		
	51		
	52		
	53		
	54		
	55		
	56		
	57		
	58		
	59		

LOGIC



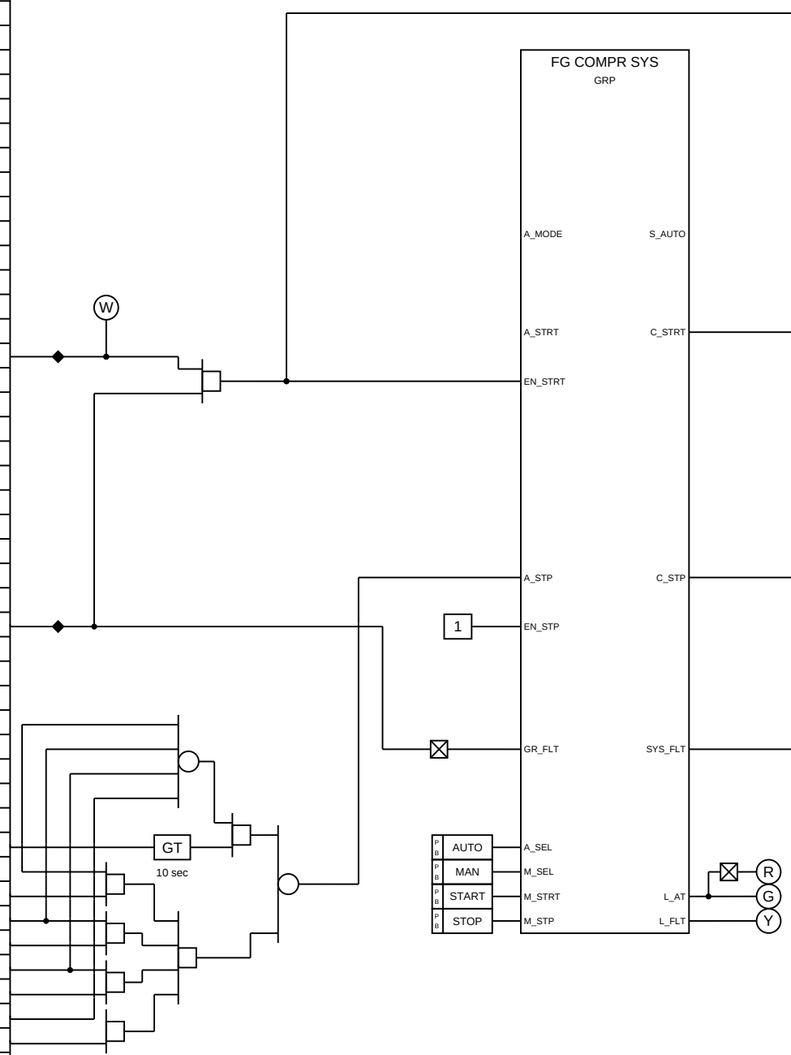
SERVICE	TAG NO.	NO.	TO
		60	
		61	
		62	
		63	
		64	
		65	
		66	
		67	
		68	
		69	
FGS GRP AUTO MODE		70	FG_G-411-10
FGS GRP AUTO MODE		71	FG_G-421-10
FGS GRP AUTO MODE		72	FG_G-431-10
FGS GRP AUTO MODE		73	FG_G-441-10
FGS GRP AUTO MODE		74	FG_G-451-10
FGS GRP AUTO MODE		75	FG_G-461-10
FGS GRP AUTO MODE		76	FG_G-511-3
FGS GRP AUTO MODE		77	FG_G-521-3
FGS GRP AUTO MODE		78	FG_G-531-3
FGS GRP AUTO MODE		79	FG_P-109-30
FGS GRP AUTO MODE		80	FG_P-110-30
FGS GRP AUTO MODE		81	FG_P-111-30
FGS GRP AUTO MODE		82	FG_G-541-10
FGS GRP AUTO MODE		83	FG_G-561-10
FGS GRP AUTO MODE		84	FG_G-581-10
FGS GRP AUTO MODE		85	FG_G-501-10
		86	
		87	
		88	
		89	
		90	
		91	
		92	
		93	
		94	
		95	
		96	
		97	
		98	
		99	
		100	
		101	
		102	
		103	
		104	
		105	
		106	
		107	
		108	
		109	
		110	
		111	
		112	
		113	
		114	
		115	
		116	
		117	
		118	

NOTE :

PROJECT NAME: AL DHAFRA POWER PLANT		
DRAWING TITLE: CONTROL LOGIC & LOOP DIAGRAM FUEL GAS SYSTEM (GROUP OPERATION)		
PROJECT DWG NO.: 1A5J-00GEN00-J09-0002	SHEET: FG_G-102	REV.: 2
PROJECT NO.: 1A5J	SCALE: NONE	REF. CODE: -

FROM	NO.	TAG NO.	SERVICE
	1		
	2		
	3		
	4		
	5		
	6		
	7		
	8		
	9		
	10		
	11		
	12		
	13		
	14		
	15		
MCP	16	00EKH01GH001XL65	FG COMPR SYS REM & RDY TO STR
	17		
	18		
	19		
	20		
	21		
	22		
	23		
	24		
	25		
	26		
MCP	27		FG COMPR SYS NO FLT
	28		
	29		
	30		
	31		
	32		
	33		
	34		
	35		
FG_P-201-68	36		GT TOTAL FLOW DEMAND = 0
GT_G-104-89	37		GTG #11 GT IN FUEL OIL OP
GT_P-102-77	38		GTG #11 ACTIV LOAD > 50%
GT_G-204-89	39		GTG #12 GT IN FUEL OIL OP
GT_P-202-77	40		GTG #12 ACTIV LOAD > 50%
GT_G-304-89	41		GTG #13 GT IN FUEL OIL OP
GT_P-302-77	42		GTG #13 ACTIV LOAD > 50%
GT_G-404-89	43		GTG #14 GT IN FUEL OIL OP
GT_P-402-77	44		GTG #14 ACTIV LOAD > 50%
	45		
MCP	46	00EKH01GH001XL01	FG COMPR IN OP
	47		
	48		
	49		
MCP	50	00EKH01GH001XL14	FG COMPR SYS COM ALM
	51		
	52		
	53		
	54		
	55		
	56		
	57		
	58		
	59		

LOGIC



SERVICE	TAG NO.	NO.	TO
		60	
FG COMPR SYS REM & RDY TO STR		61	FG_G-101-13
		62	
		63	
		64	
		65	
		66	
		67	
		68	
		69	
		70	
		71	
		72	
		73	
FG COMPR SYS GRP STR CMD	00EKH01GH001YL01	74	MCP
		75	
		76	
		77	
		78	
		79	
		80	
		81	
		82	
		83	
FG COMPR SYS GRP STP CMD	00EKH01GH001YL02	84	MCP
		85	
		86	
		87	
		88	
		89	
		90	
FG COMPR SYS FAULT		91	
		92	
		93	
		94	
		95	
		96	
		97	
		98	
		99	
		100	
		101	
FG COMPR IN OP		102	GT_G-111-12
FG COMPR IN OP		103	GT_G-211-12
FG COMPR IN OP		104	GT_G-311-12
FG COMPR IN OP		105	GT_G-411-12
		106	
		107	
		108	
FG COMPR SYS COM ALM		109	HMI
		110	
		111	
		112	
		113	
		114	
		115	
		116	
		117	
		118	

NOTE :

PROJECT NAME	AL DHAFRA POWER PLANT		
DRAWING TITLE	CONTROL LOGIC & LOOP DIAGRAM FG COMPR SYS (MCP)		
PROJECT DWG NO.	1A5J-00GEN00-J09-0002	SHEET	FG_G-201
		REV.	2
PROJECT NO.	1A5J	SCALE	NONE
		REF. CODE	-

FROM	NO.	TAG NO.	SERVICE	LOGIC	SERVICE	TAG NO.	NO.	TO
	1						60	
	2						61	
	3			G			62	
MCP	4	00EKH01AN001XL01	FG COMPR #1 RUN		FG COMPR # 1 RUN		63	HMI
	5						64	
	6						65	
	7			G			66	
MCP	8	00EKH02AN001XL01	FG COMPR #2 RUN		FG COMPR # 2 RUN		67	HMI
	9						68	
	10						69	
	11			G			70	
MCP	12	00EKH03AN001XL01	FG COMPR #3 RUN		FG COMPR # 3 RUN		71	HMI
	13						72	
	14						73	
	15			G			74	
MCP	16	00EKH04AN001XL01	FG COMPR #4 RUN		FG COMPR # 4 RUN		75	HMI
	17						76	
	18						77	
	19			G			78	
MCP	20	00EKH05AN001XL01	FG COMPR #5 RUN		FG COMPR # 5 RUN		79	HMI
	21						80	
	22			ANN			81	
	23						82	
MCP	24	00EKH01AN001XL02	FG COMPR #1 TRIP		FG COMPR # 1 TRIP		83	HMI
	25						84	
	26			ANN			85	
	27						86	
MCP	28	00EKH02AN001XL02	FG COMPR #2 TRIP		FG COMPR # 2 TRIP		87	HMI
	29						88	
	30			ANN			89	
	31						90	
MCP	32	00EKH03AN001XL02	FG COMPR #3 TRIP		FG COMPR # 3 TRIP		91	HMI
	33						92	
	34			ANN			93	
	35						94	
MCP	36	00EKH04AN001XL02	FG COMPR #4 TRIP		FG COMPR # 4 TRIP		95	HMI
	37						96	
	38			ANN			97	
	39						98	
MCP	40	00EKH05AN001XL02	FG COMPR #5 TRIP		FG COMPR # 5 TRIP		99	HMI
	41						100	
	42						101	
	43						102	
	44						103	
	45						104	
	46						105	
	47						106	
	48						107	
	49						108	
	50						109	
	51						110	
	52						111	
	53						112	
	54						113	
	55						114	
	56						115	
	57						116	
	58						117	
	59						118	

NOTE :

PROJECT NAME	AL DHAFRA POWER PLANT		
DRAWING TITLE	CONTROL LOGIC & LOOP DIAGRAM FG COMPR SYS #1		
PROJECT DWG NO.	1A5J-00GEN00-J09-0002	SHEET	FG_G-202
		REV.	2
PROJECT NO.	1A5J	SCALE	NONE
		REF. CODE	-

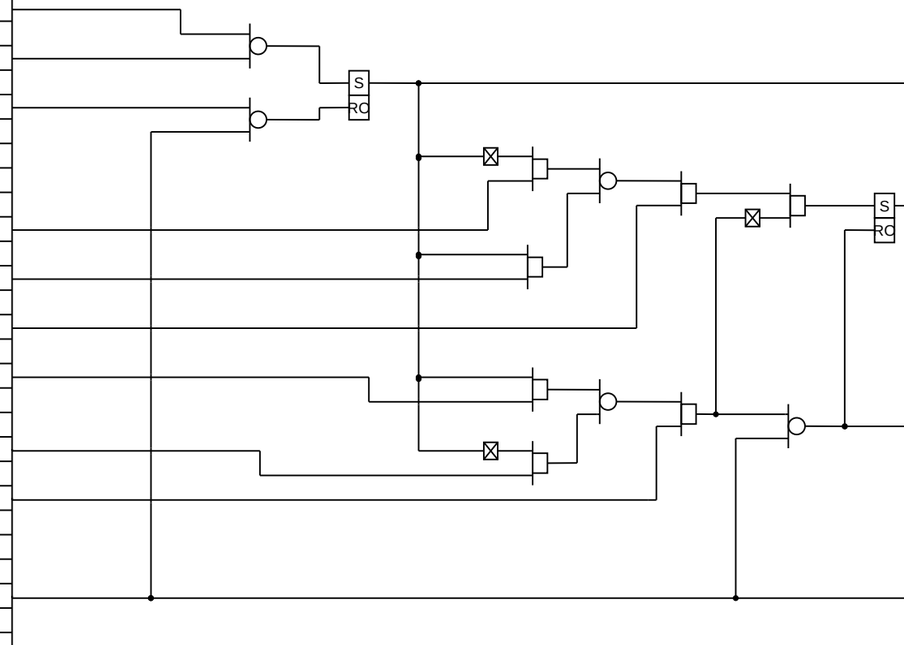
FROM	NO.	TAG NO.	SERVICE	LOGIC	SERVICE	TAG NO.	NO.	TO
	1						60	
	2						61	
	3						62	
	4						63	
	5						64	
	6						65	
	7						66	
	8						67	
GT_G-105-65	9		GTG #11 IN OP		GTG #11 RUN	00EKH01GH001YL54	68	MCP
	10						69	
	11						70	
	12						71	
GT_G-205-65	13		GTG #12 IN OP		GTG #12 RUN	00EKH01GH001YL55	72	MCP
	14						73	
	15						74	
	16						75	
GT_G-305-65	17		GTG #13 IN OP		GTG #13 RUN	00EKH01GH001YL56	76	MCP
	18						77	
	19						78	
	20						79	
GT_G-405-65	21		GTG #14 IN OP		GTG #14 RUN	00EKH01GH001YL57	80	MCP
	22						81	
	23						82	
	24						83	
GT_G-105-77	25		GTG #11 IN TRIPPED CONDITION		GTG #11 TRIPD	00EKH01GH001YL41	84	MCP
	26						85	
	27						86	
	28						87	
GT_G-103-83	29		GTG #11 IN LOAD REJ CHAN A		GTG #11 LOAD REJ	00EKH01GH001YL42	88	MCP
GT_G-103-89	30		GTG #11 IN LOAD REJ CHAN B				89	
	31						90	
	32						91	
GT_G-205-77	33		GTG #12 IN TRIPPED CONDITION		GTG #12 TRIPD	00EKH01GH001YL43	92	MCP
	34						93	
	35						94	
	36						95	
GT_G-203-83	37		GTG #12 IN LOAD REJ CHAN A		GTG #12 LOAD REJ	00EKH01GH001YL44	96	MCP
GT_G-203-89	38		GTG #12 IN LOAD REJ CHAN B				97	
	39						98	
	40						99	
GT_G-305-77	41		GTG #13 IN TRIPPED CONDITION		GTG #13 TRIPD	00EKH01GH001YL45	100	MCP
	42						101	
	43						102	
	44						103	
GT_G-303-83	45		GTG #13 IN LOAD REJ CHAN A		GTG #13 LOAD REJ	00EKH01GH001YL46	104	MCP
GT_G-303-89	46		GTG #13 IN LOAD REJ CHAN B				105	
	47						106	
	48						107	
GT_G-405-77	49		GTG #14 IN TRIPPED CONDITION		GTG #14 TRIPD	00EKH01GH001YL47	108	MCP
	50						109	
	51						110	
	52						111	
GT_G-403-83	53		GTG #14 IN LOAD REJ CHAN A		GTG #14LOAD REJ	00EKH01GH001YL48	112	MCP
GT_G-403-89	54		GTG #14 IN LOAD REJ CHAN B				113	
	55						114	
	56						115	
	57						116	
	58						117	
	59						118	

NOTE :

PROJECT NAME: AL DHAFRA POWER PLANT		
DRAWING TITLE: CONTROL LOGIC & LOOP DIAGRAM FG COMPR INTERFACE		
PROJECT DWG NO.: 1A5J-00GEN00-J09-0002	SHEET: FG_G-207	REV.: 2
PROJECT NO.: 1A5J	SCALE: NONE	REF. CODE: -

FROM	NO.	TAG NO.	SERVICE
	1		
	2		
	3		
	4		
	5		
	6	A_SEL	AUTO SELECT
	7		
	8	A_MODE	EXT AUTO MODE REQUEST
	9		
	10	M_SEL	MANUAL SELECT
	11		
	12		
	13		
	14		
	15	M_STRT	MANUAL START REQUEST
	16		
	17	A_STRT	AUTO START REQUEST
	18		
	19	EN_STRT	START PERMISSIVE CONDITION
	20		
	21	A_STP	AUTO STOP REQUEST
	22		
	23		
	24	M_STP	MANUAL STOP REQUEST
	25		
	26	EN_STP	STOP PERMISSIVE CONDITION
	27		
	28		
	29		
	30	GR_FLT	DRIVE GROUP FAULT
	31		
	32		
	33		
	34		
	35		
	36		
	37		
	38		
	39		
	40		
	41		
	42		
	43		
	44		
	45		
	46		
	47		
	48		
	49		
	50		
	51		
	52		
	53		
	54		
	55		
	56		
	57		
	58		
	59		

LOGIC



SERVICE	TAG NO.	NO.	TO
		60	
		61	
		62	
		63	
		64	
		65	
		66	
AUTO MODE	L_AT	67	
SYS/GRP AUTO MODE	S_AUTO	68	
		69	
		70	
		71	
		72	
SYS/GRP START COMMAND	C_STRT	73	
		74	
		75	
		76	
		77	
		78	
		79	
		80	
		81	
SYS/GRP STOP COMMAND	C_STP	82	
		83	
		84	
		85	
		86	
		87	
		88	
SYSTEM FAULT	SYS_FLT	89	
SYSTEM FAULT	L_FLT	90	
		91	
		92	
		93	
		94	
		95	
		96	
		97	
		98	
		99	
		100	
		101	
		102	
		103	
		104	
		105	
		106	
		107	
		108	
		109	
		110	
		111	
		112	
		113	
		114	
		115	
		116	
		117	
		118	

NOTE :

PROJECT NAME	AL DHAFRA POWER PLANT	
DRAWING TITLE	CONTROL LOGIC & LOOP DIAGRAM MACRO GRP <GROUP OPERATION>	
PROJECT DWG NO. 1A5J-00GEN00-J09-0002	SHEET 0012	REV. 2
PROJECT NO. 1A5J	SCALE NONE	REF. CODE -